

Different technologies in the gas sensing market

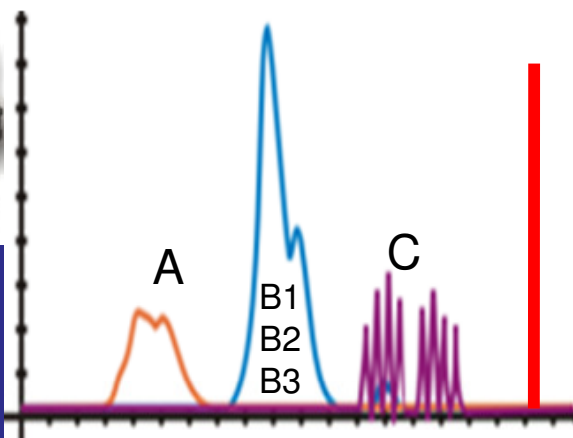
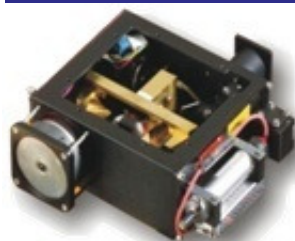
FTIR



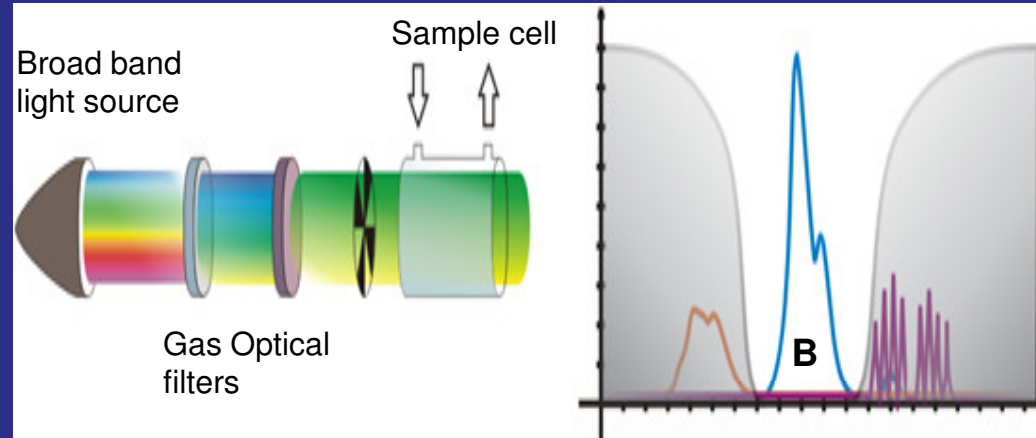
- 1) NDIR
- 2) Electrochemical sensor
- 3) Manual sampling
(not in real time)

Technologies - FTIR VS NDIR/ GFC

Fourier Transformation Infrared (Gasmet)



Non-Dispersive Infrared/ Gas Correlation Filter (Others)



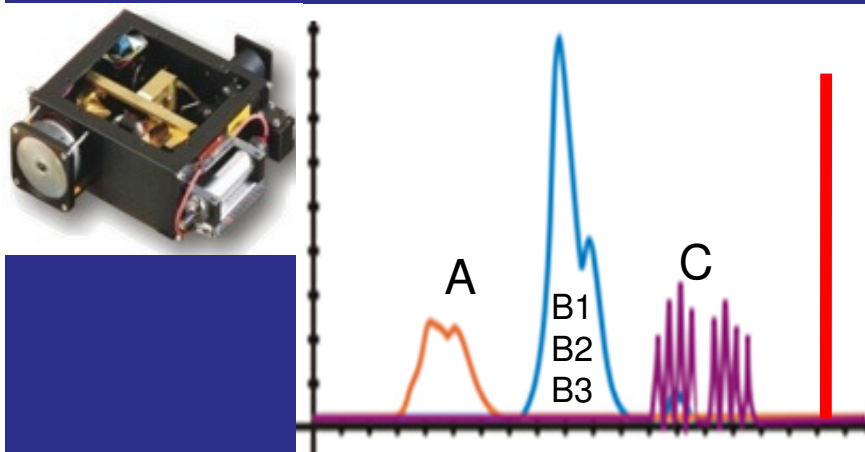
- From normal stack gas to complicated VOC gas matrix
- Expandable to 50 gases
- Water and CO₂ plus other gas cross - interference correction
- Use N₂ gas for calibration only
- Gas reference is stable

Verstaile, good accuracy and low running cost

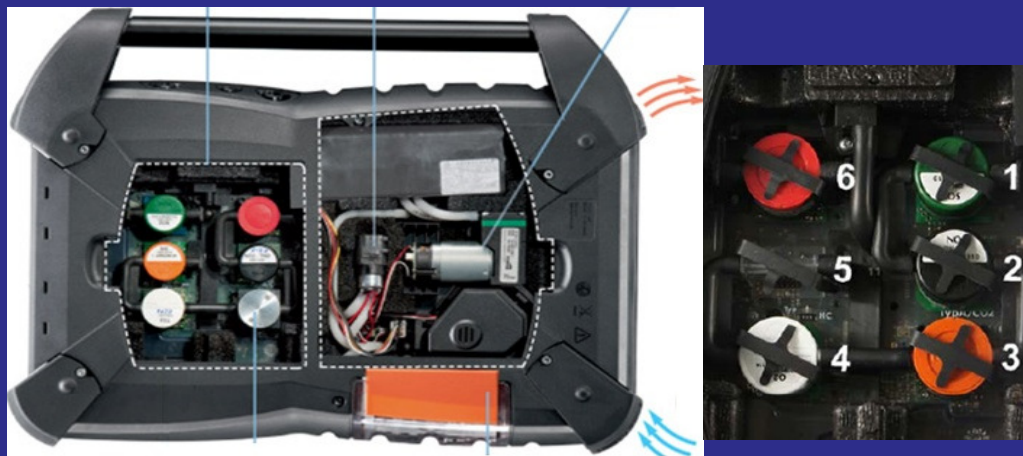
- Measures only narrow wavelength bands with gas-filled filters
- Cross sensitive to H₂O, difficult to analyze low SO₂/ NO₂/HCl/ HF concentration
→ Less accurate (**results not specific**)
- Only one component can be measured for each filter, **maximum of 6 filters** (available SO₂/CO/CO₂)
- Need to combined with CLD for NO/NO₂ or electrochemical sensors
- **Multi-gas calibration are needed** for maintaining it's stability (weekly) → Costly

Technologies - FTIR VS Electrochemical sensor

Fourier Transformation Infrared (Gasmet)



Electrochemical sensors (Others)

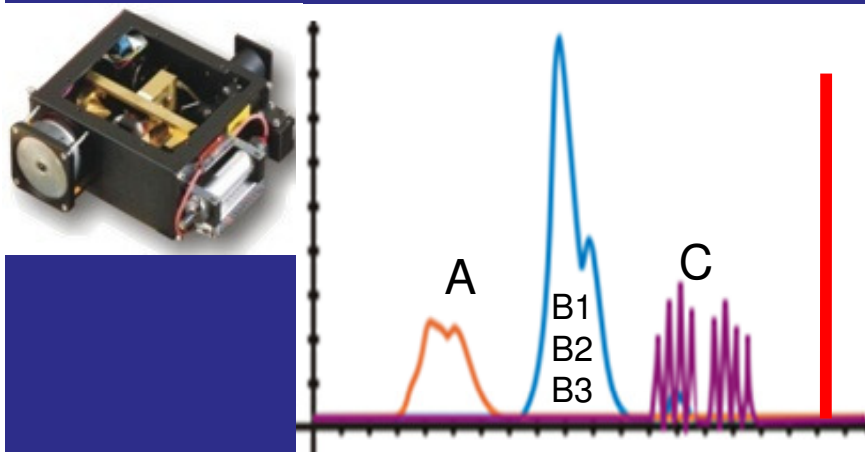


- From normal stack gas to complicated VOC gas matrix
 - Expandable to 50 gases
 - Water and CO₂ plus other gas cross-interference correction
 - Use N₂ gas for calibration only
 - Gas reference is stable
- Verstaile, good accuracy and low running cost

- Limited to NO_x/SO₂/CO/O₂/H₂S/few HCs
- Need to combine with NDIR for CO₂ monitoring
- Suffer from serious cross interference problem
→ less accurate (results not specific)
- Very short lifetime (~half to 1 year), especially in wet and hot areas. **Gas sensors are not under warranty, high running cost**
- Gas sensors must be replaced as a whole and in dedicated slot

Technologies - FTIR VS Electrochemical sensor

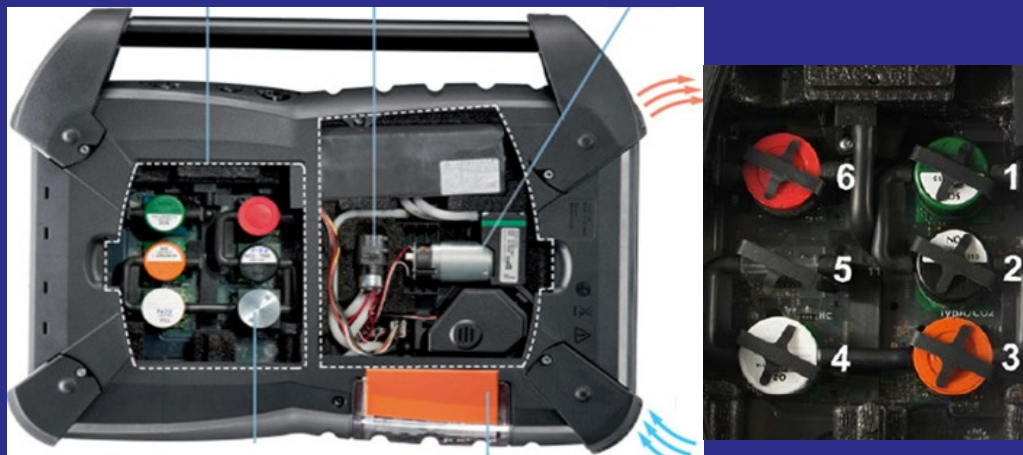
Fourier Transformation Infrared (Gasmet)



- From normal stack gas to complicated VOC gas matrix
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Verstaile, good accuracy and low running cost

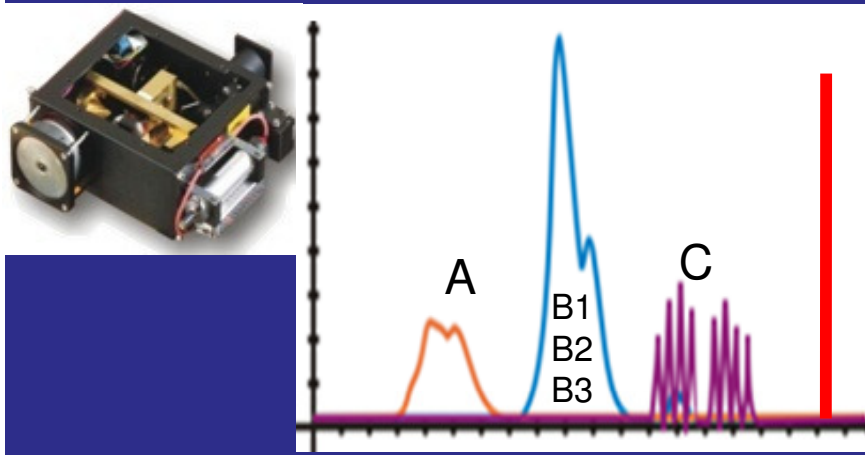
Electrochemical sensors (Others)



- **Multi-gas calibration is needed** for maintaining it's stability (monthly) → Costly
- Weak sample pump ~1.0 lpm is usually used, longer response time

Technologies - FTIR VS Manual Sampling

Fourier Transformation Infrared (Gasmet)



Manual Sampling → Analytical Instruments in Lab



- FTIR proven Technology: become more popular for any testing standard nowadays
- System can be operated by person with relatively short training
- Standard alone method, no supporting lab facilities are needed.
- Analyze samples in real time and continuously
- Quantitative and qualitative, can be used for organic and inorganic components

- Methodology varies according to the standard applied
- Special Sampling technique may required (isokinetic sampling, Canister, test tube, Tedlar bag)
 - Sample may lost during transportation
- All necessary analyzers e.g. GC, must be equipped in laboratory
- **Not a real time** data, may takes one day for analysis. (3rd party on-site test would be an issue)
- May be sensitive (ppb), but not quantitatively accurate.
- Expensive (time and labor based) and slow response